
Other AI Fields

INSTRUCTOR: MOAYAD ALNAMMI

malnammi.github.io

Notice

- Slides prepared by Moayad Alnammi – malnammi.github.io
- Some figures adapted from CS188: Introduction to Artificial Intelligence, UC Berkeley (Dan Klein, Pieter Abbeel) — ai.berkeley.edu
- Some figures are obtained from the [AIMA Book](#)

This deck is published at malnammi.github.io under CC BY-NC-SA 4.0. Reuse and adaptation require attribution to the author and the sources above; commercial use is not permitted; adaptations must carry the same license.

Outline

- Recap of what we covered
- Overview of what we did not cover:
 - Deep learning
 - NLP
 - Robotics

Rough overview of what we covered

- **Planning/Searching:** given known environment and some destination, **computationally search** and propose a path.
- **Local Search and Constraint-satisfaction (optimization):** given an environment and some conditions, **computationally search** and find a state of the environment that satisfies the condition.
- **Game playing:** how to operate in 2-player (or multiplayer) environments.
- **Knowledge-based agents:** develop agents that can answer questions about the environment it is in based on the knowledge (stored in it) it has so far. This includes logic-based and probabilistic-based agents.

Navigation & Searching

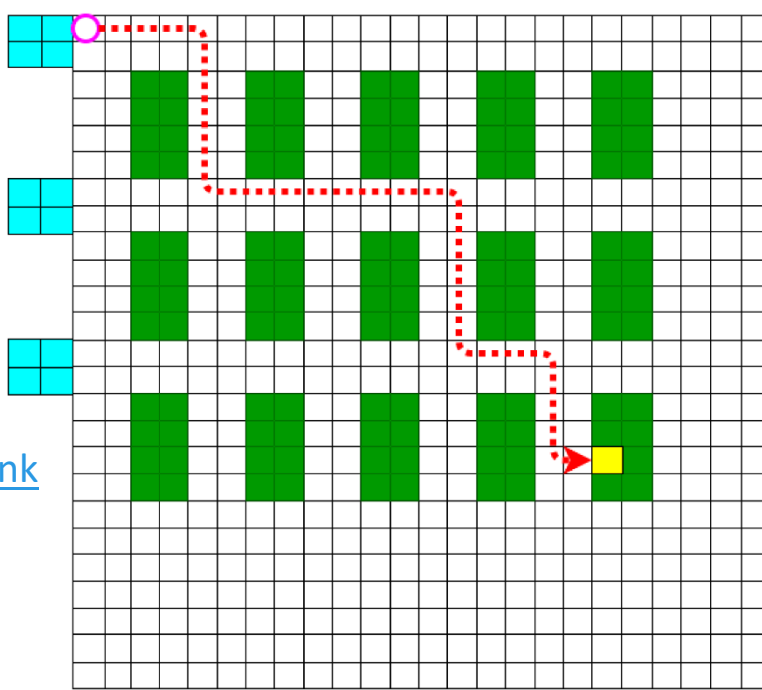


Image from: [link](#)

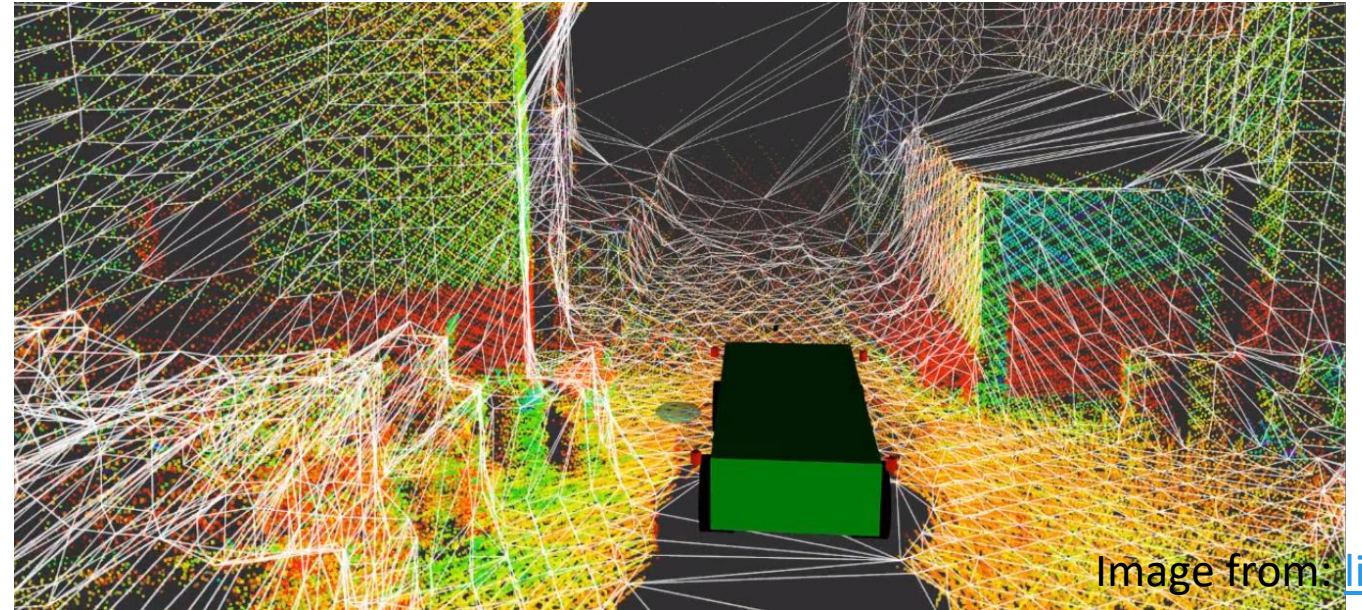
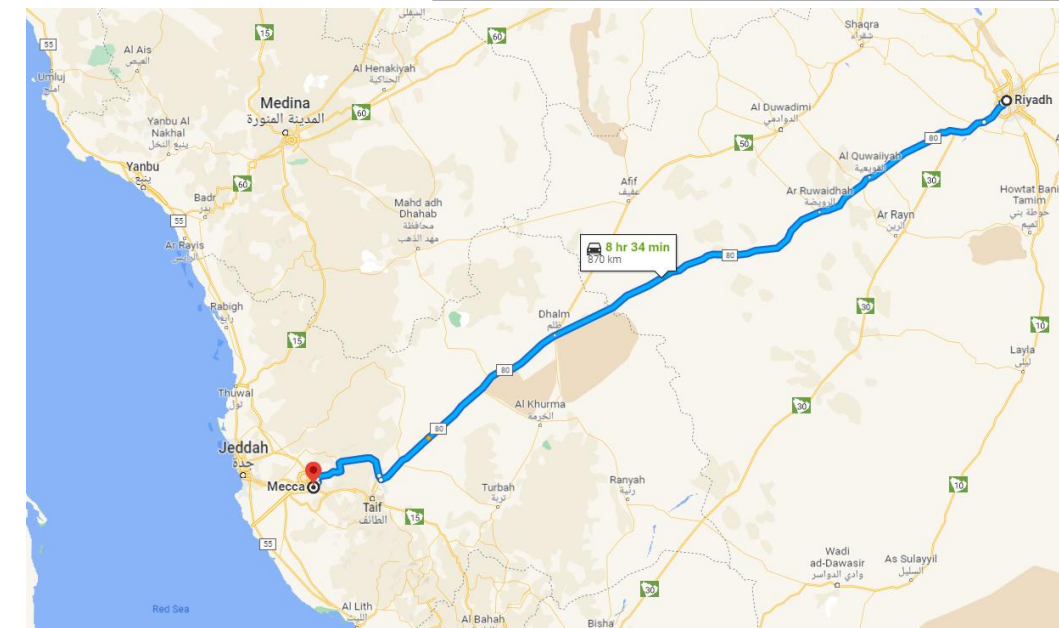


Image from: [link](#)

Scheduling/Optimizing

- Finding optimal configuration of a system.
- Job scheduling
- Circuit design
- Cargo packing
- Flight scheduling
- Registrar scheduling

Course Offering

Select Term

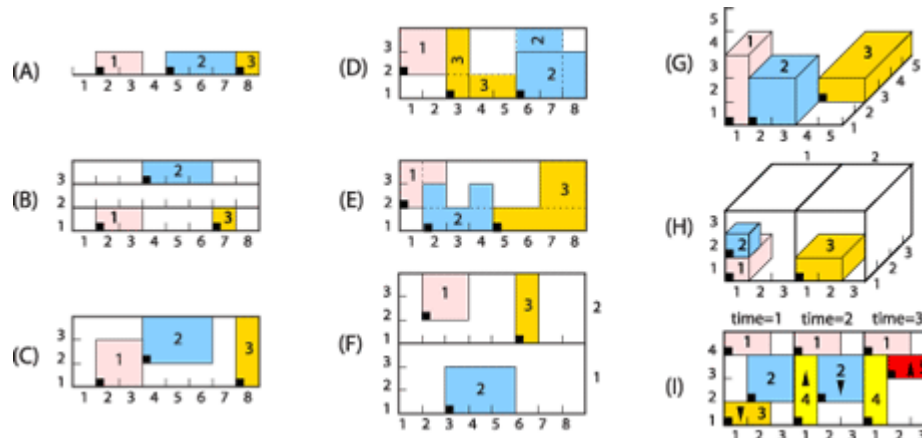
Term 222

Select Department

Accounting & Finance

Course-Sec	Activity	CRN	Course Name	Instructor	Day	Time	Location	Status	Waitlist
ACCT110-01	LEC	21684	Introduction to Financial Accounting	ABDULLAH ALHARBI	MW	1230-1345	24-174	Closed	Open
ACCT110-02	LEC	21083	Introduction to Financial Accounting	ABDULLAH ALHARBI	MW	1400-1515	24-174	Open	Closed
ACCT210-01	LEC	21087	Introduction to Managerial Accounting	MIRGHANI AHMED	UT	1100-1215	24-174	Closed	Open
ACCT210-04	LEC	21090	Introduction to Managerial Accounting	WALID CHEFFI	MW	1230-1345	24-244	Open	Closed
ACCT210-05	LEC	21982	Introduction to Managerial Accounting	MIRGHANI AHMED	UT	1400-1515	24-250	Open	Closed

Course-instructor-room scheduling



Packing problems. Image from [link](#)

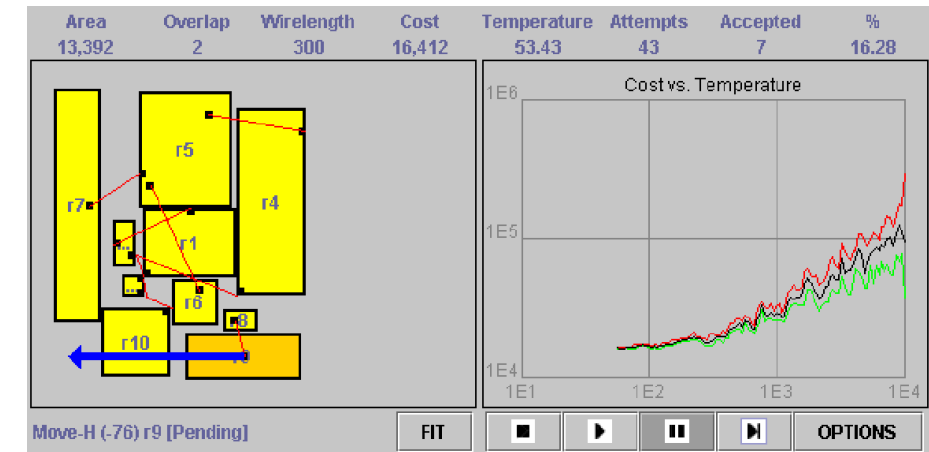
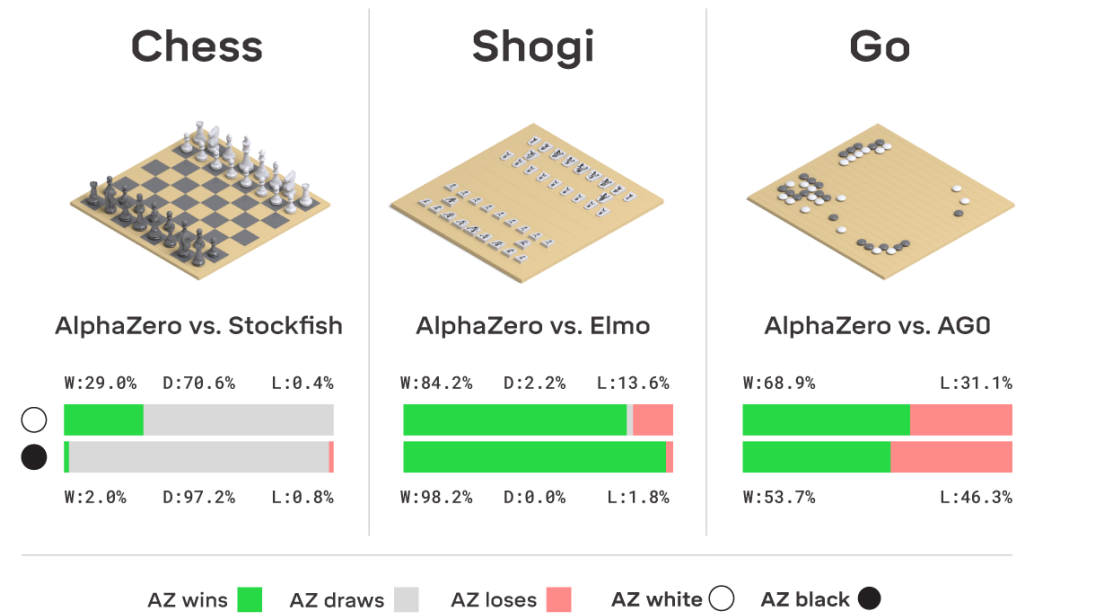


Figure 5 – Simulated Annealing Placement Applet

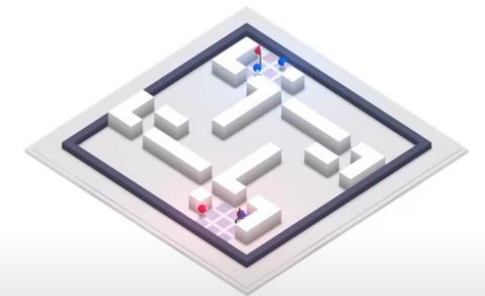
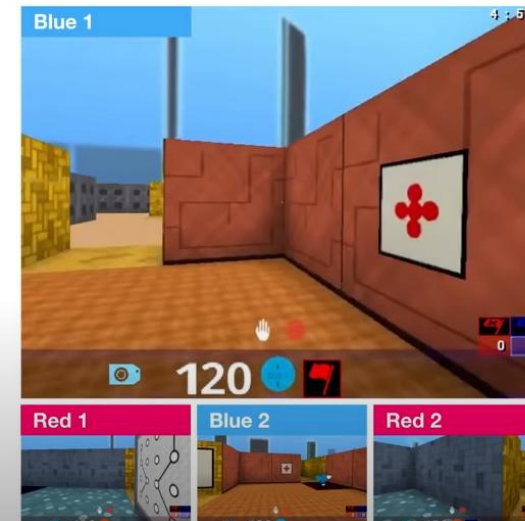
Circuit design. Image from [link](#)

Game Playing

- DeepBlue defeats chess champion [1997]
- AlphaZero defeats humans and other AI in chess, shogi and go [2018].
- Open AI Five beats Dota champions. [2018]
- [Deepmind's Quake 3 AI](#) [2018].
- [AlphaStar](#): Grandmaster level in StarCraft II [2019].



Agent observation raw pixels



- Search Engines
- Recommender systems
- Spam Filters
- Automated helpdesks
- Medical diagnosis systems
- Automated trading
- Fraud Detection
- Fault Detection

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 B OK	2,2	3,2	4,2
1,1 OK	2,1 B OK	3,1	4,1

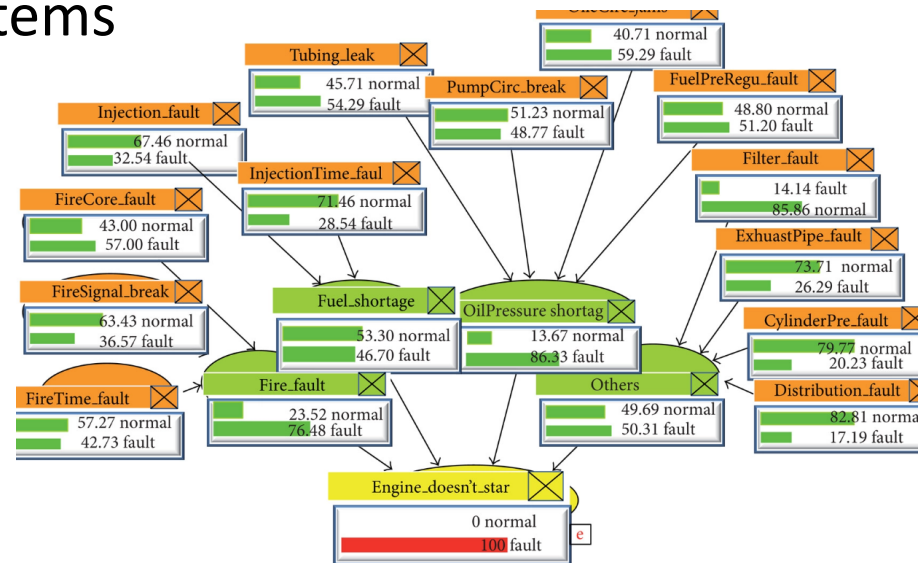
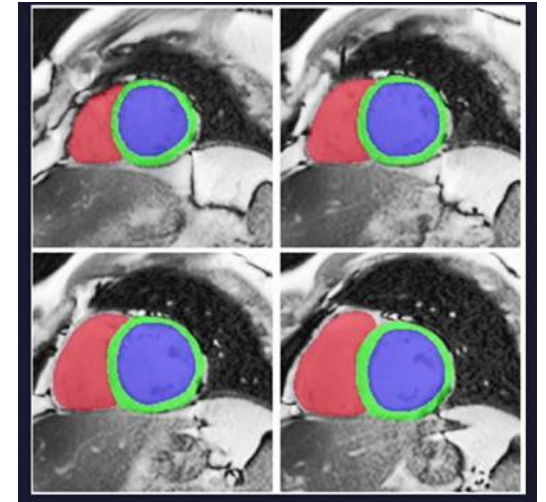


Image from [link](#)

amazon.com

Recommended for You

Amazon.com has new recommendations for you based on [items](#) you purchased or told us you own.



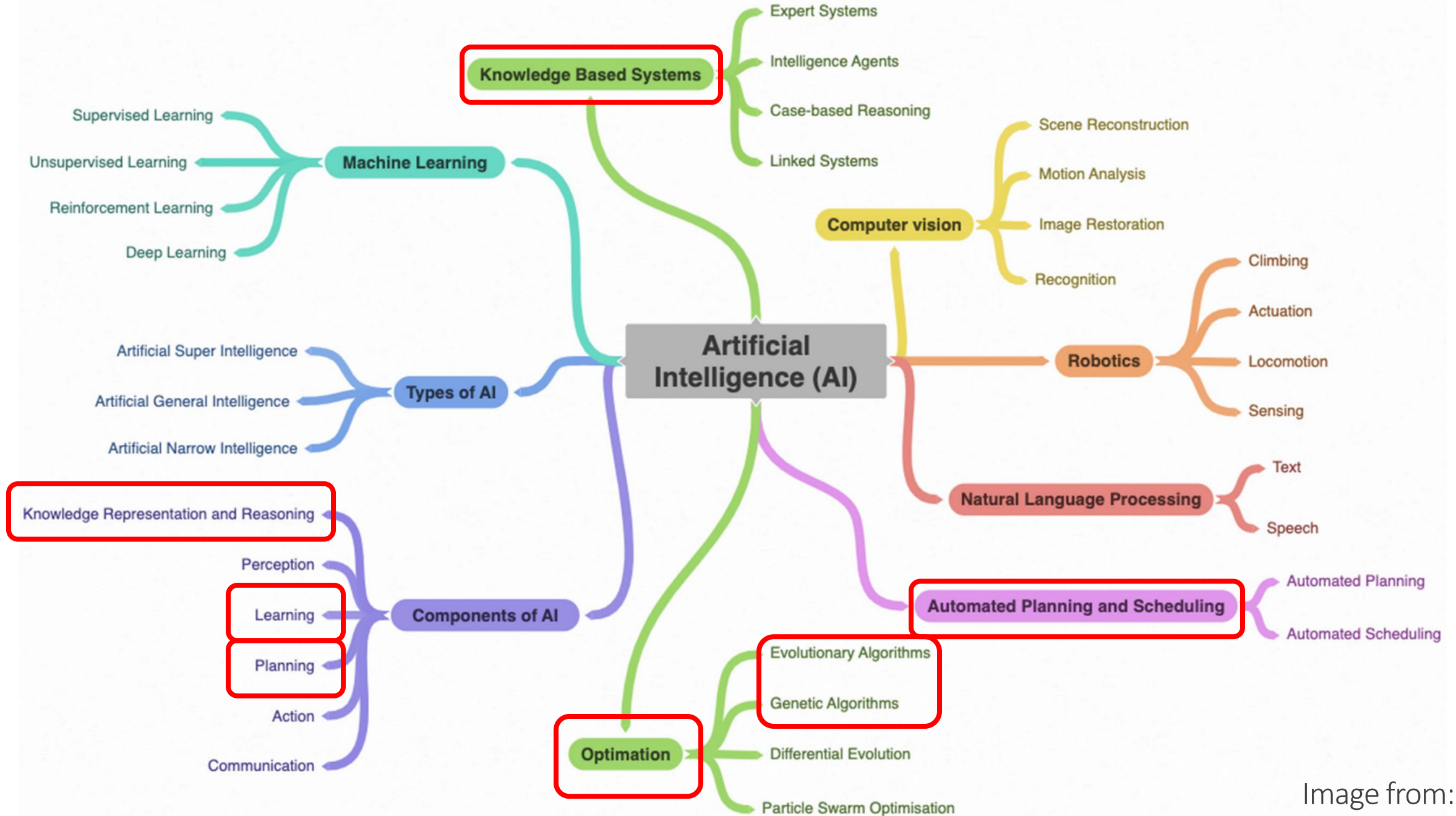


Image from: [link](#)

Rough overview of what we did not cover

- **Deep learning:** concerned with machine learning algorithms based on trainable graphs. Popular approach for image/video data, audio data, and language-based data (e.g. ChatGPT).
- **Natural Language Processing (NLP):** concerned with algorithms for processing/analyzing speech and textual data.
- **Robotics:** concerned with algorithms for executing actions at the physical level (control, sensors, actuators).

Deep Learning

History of Deep Learning [Neural Networks]

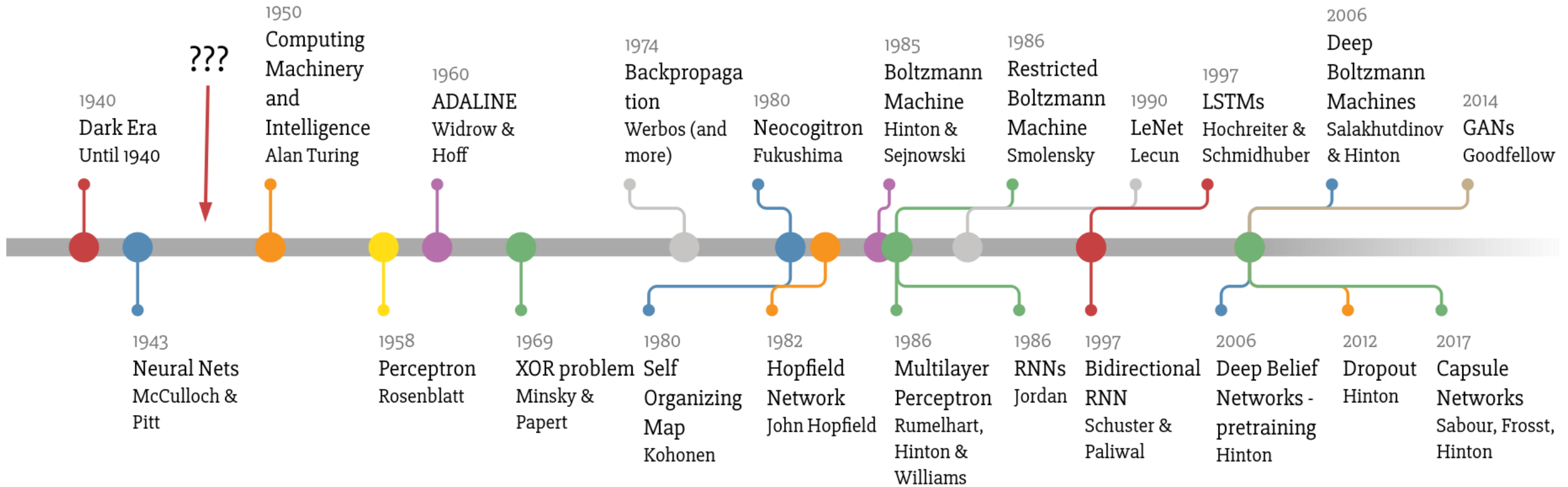
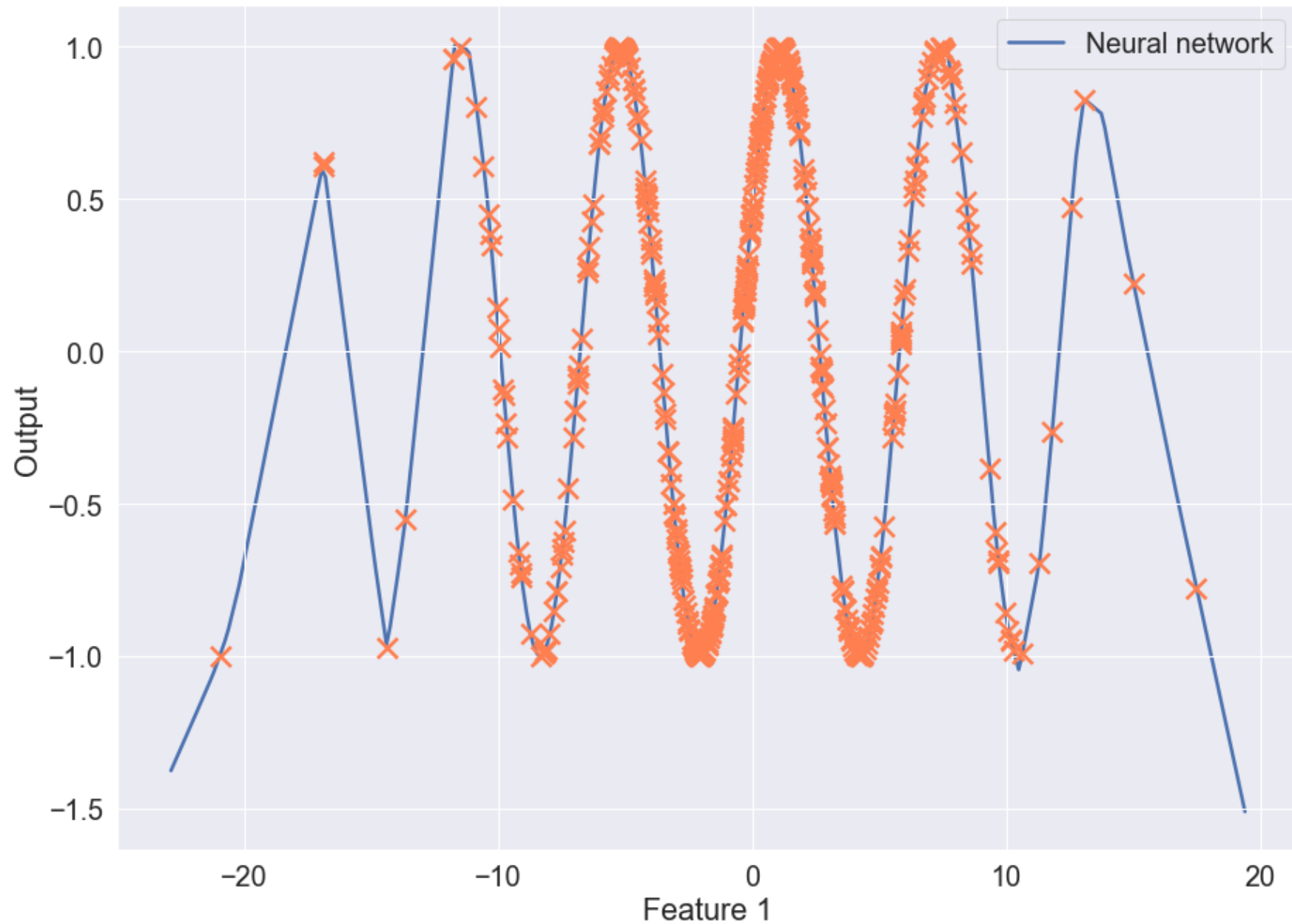


Image from: [link](#)

Neural Networks

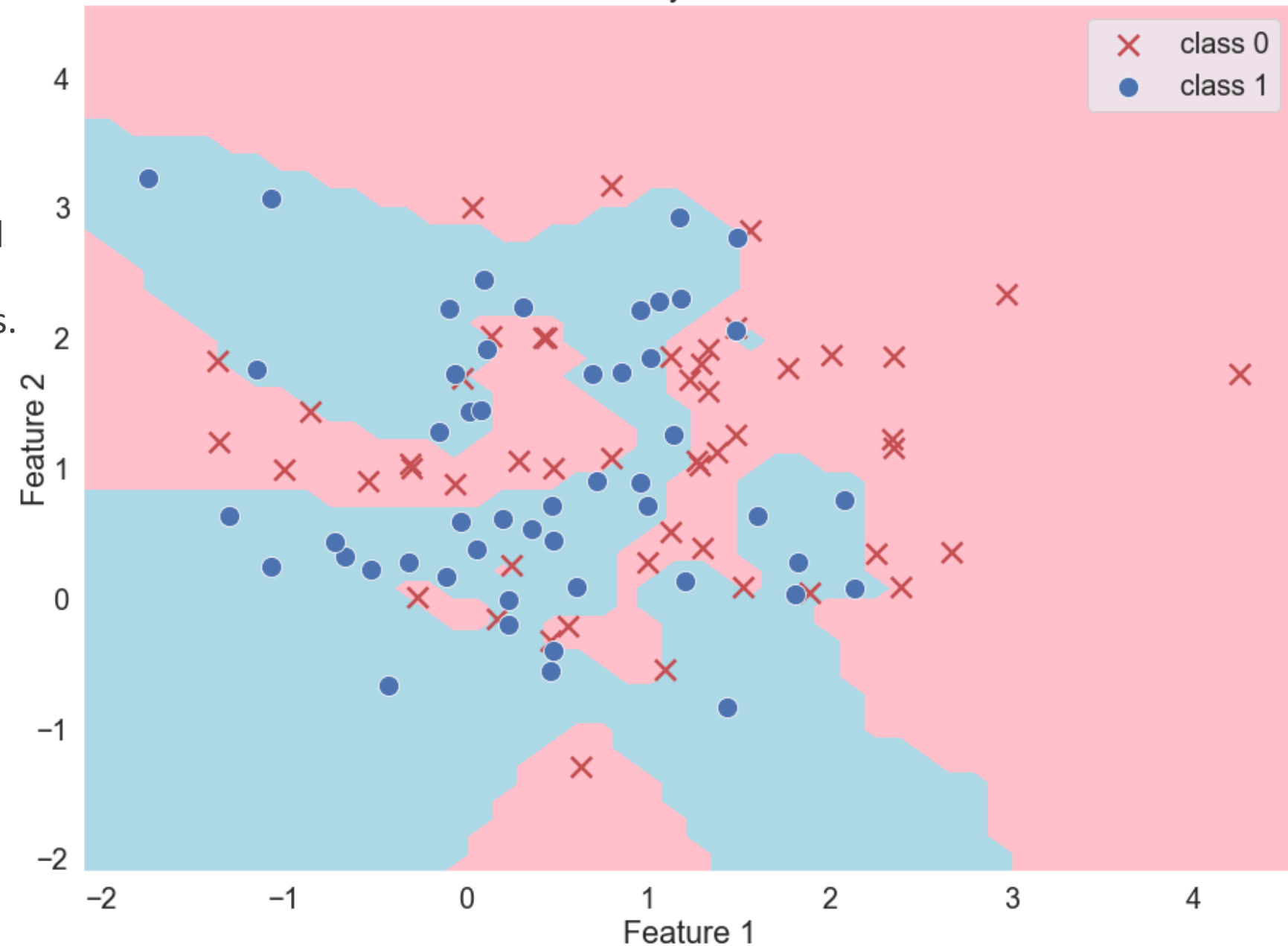
- Inspired by biological networks in the brain. Brain neurons are interconnected via synapses. These neurons communicate with each via these synapse connections.
- In neural networks, we represent the neurons as **nodes** and the synapses\connections as **weights**.
- In other words, a neural network is represented as a directed graph. Nodes point to other nodes via arrows. The arrows indicate the direction of data flow.
- Very powerful ML algorithms capable of approximating complex functions. Thus, with lots of data, leads to a powerful trained model.
- Due to the internet, advancements in data centers, advancements in machines (particularly GPU devices), this has lead to resurgence of neural networks since the 2000s.

- See the one-feature regression dataset. A neural network was fit to this training data.
- Note that the network takes in feature 1 as input. So we did not add any polynomial transformations of feature 1.
- Neural networks are capable of learning complex functions without the need to transform the input features beforehand like with polynomial linear regression.

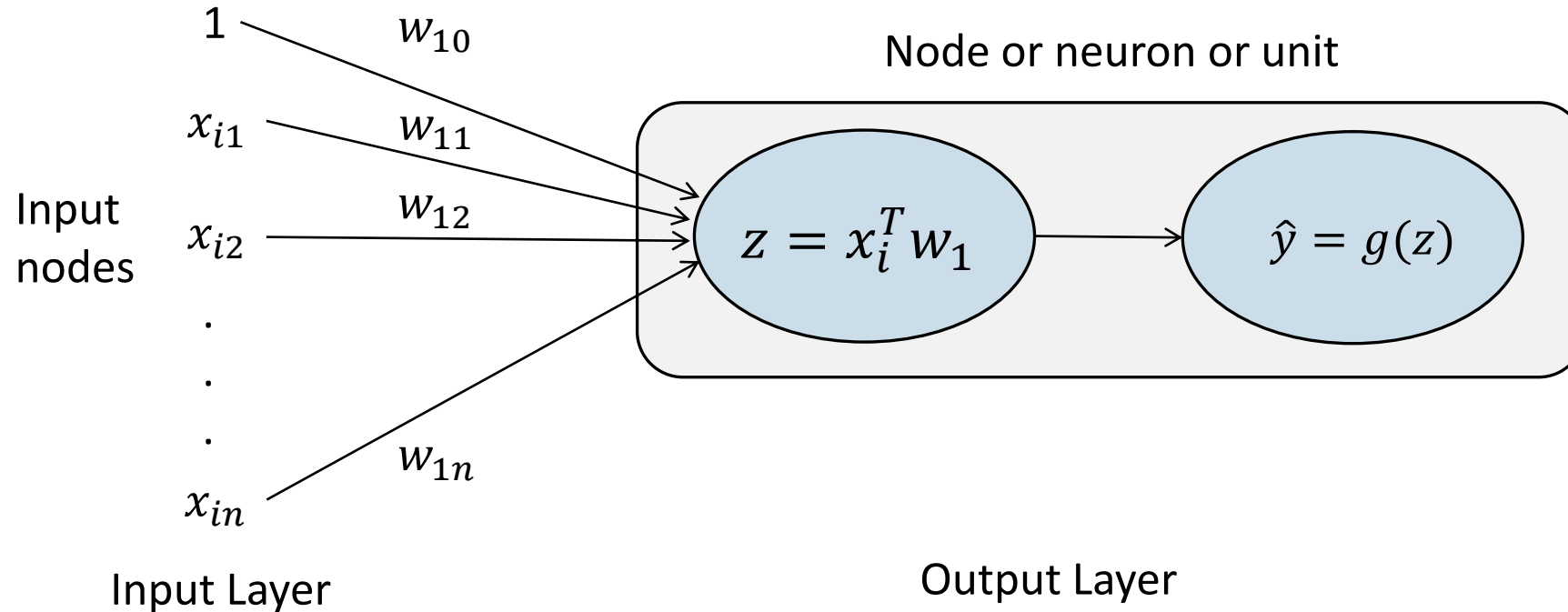


Neural network fit for a binary classification two-feature task

- Two-feature binary classification dataset. A neural network was fit to this training data.
- Note that the network takes in the two features as input. We did not add any polynomial transformations of these features.



Single Layer neural networks



In general, you can apply any function g for a node. This is called the **activation function**. The only requirement is that it is differentiable (needed for training method used). Also recommended to be fast to compute.

What would g be to mimic linear regression? What about logistic regression?

Multilayer NN

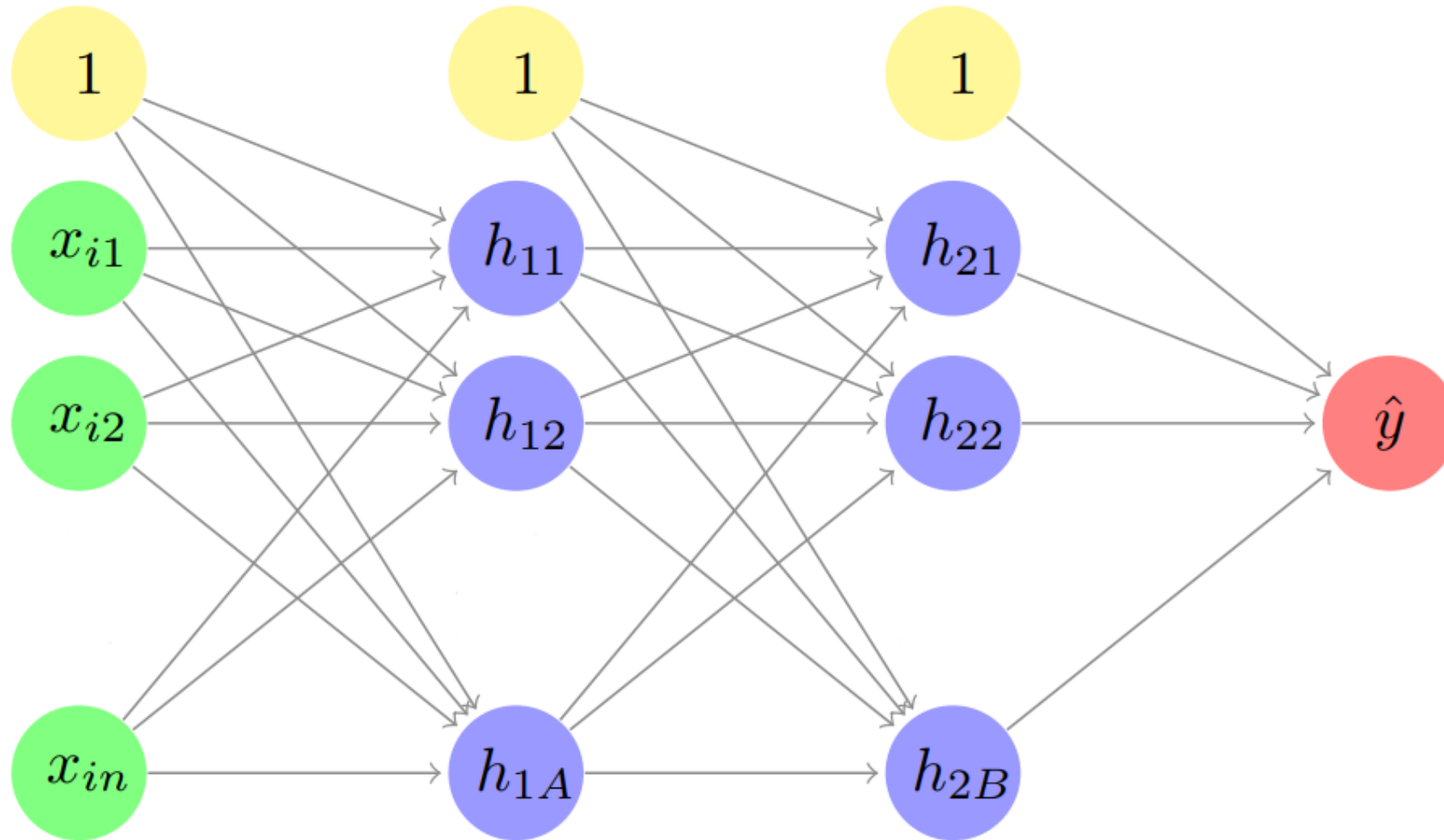
- A single layer network is not very powerful. It is just ordinary linear or logistic regression. What can we do to increase its complexity?
- Add intermediate layers between the input layer and output layer. These intermediate layers are called **Hidden Layers**.

Input
layer 1

Hidden
layer 1

Hidden
layer 2

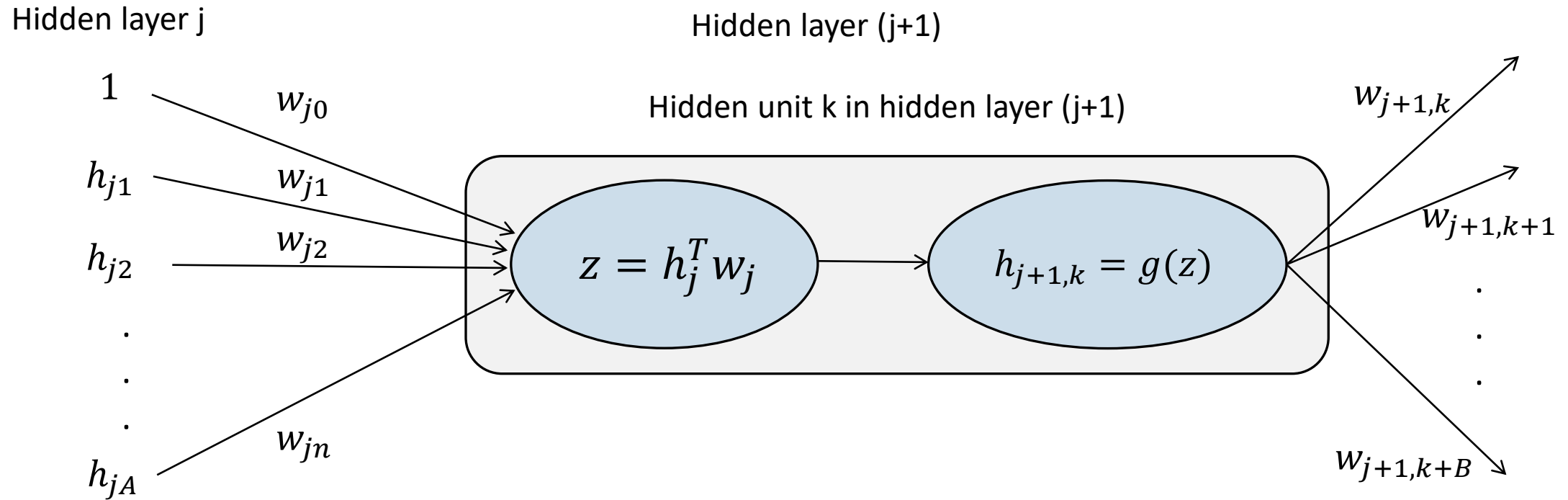
Output
layer



- This is a multilayer network with 2 hidden layers.
- Each arrow represents a weight value that needs to be learned via training.
- Each hidden layer contains nodes that take in the weighted sum from the previous layer.
- Notice that hidden layers can have different number of units.
- The hidden units perform an activation function like sigmoid or ReLu.
- The final output layer is the prediction.
- Deep learning -> having many hidden layers.

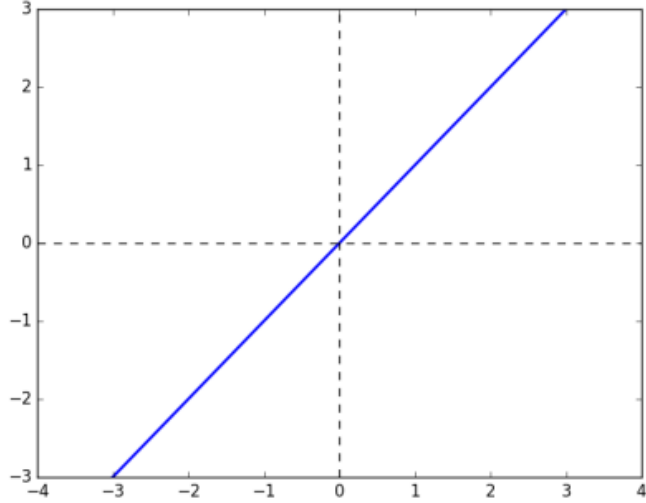
Multilayer NN [2]

- Each hidden unit takes in the weighted sum of the previous layer and performs an activation function.



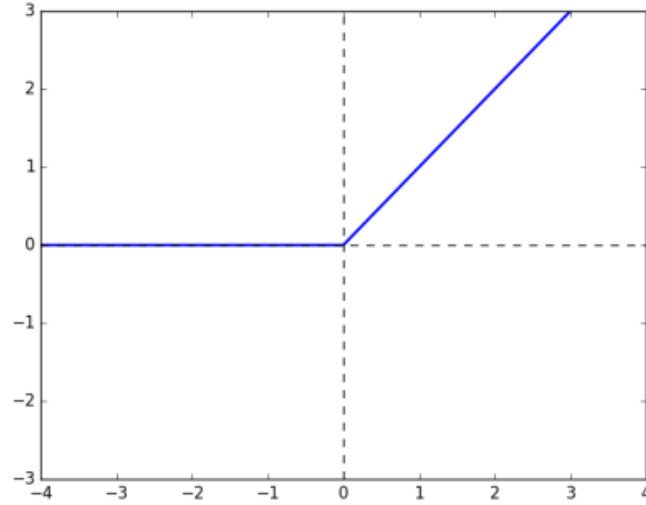
- Using **non-linear** activation functions allows the network to approximate highly complex non-linear functions.

Common activation function choices [1]



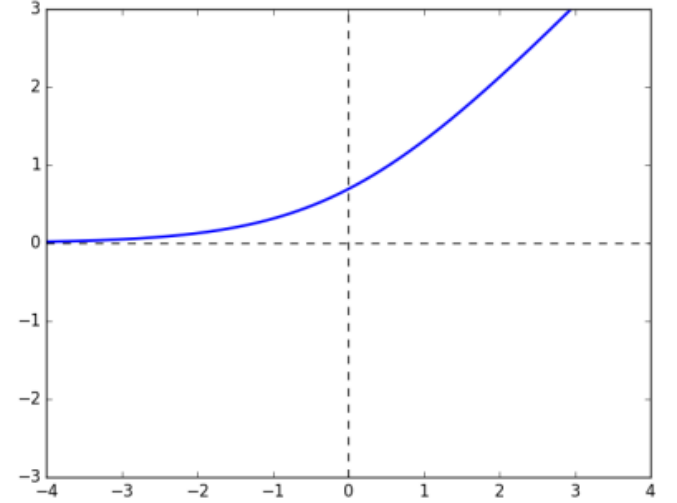
Linear

$$y = z$$



**Rectified Linear Unit
(ReLU)**

$$y = \max(0, z)$$

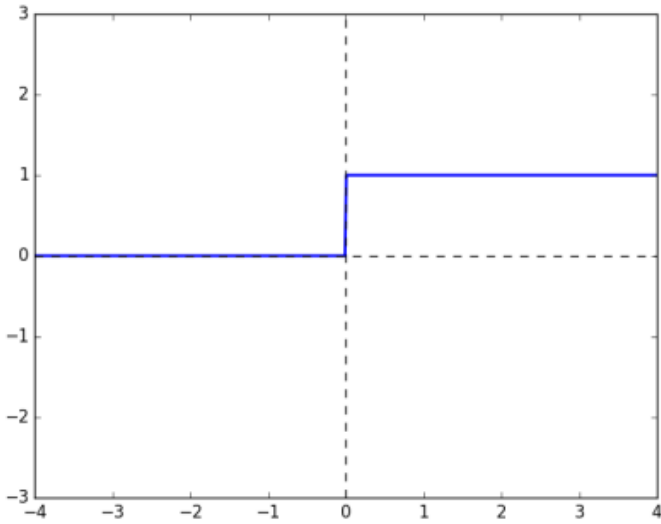


Soft ReLU

$$y = \log 1 + e^z$$

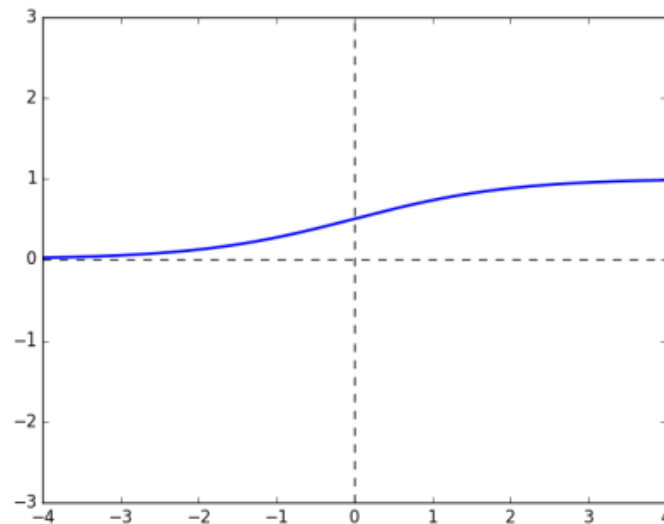
Image from: [link](#)

Common activation function choices [2]



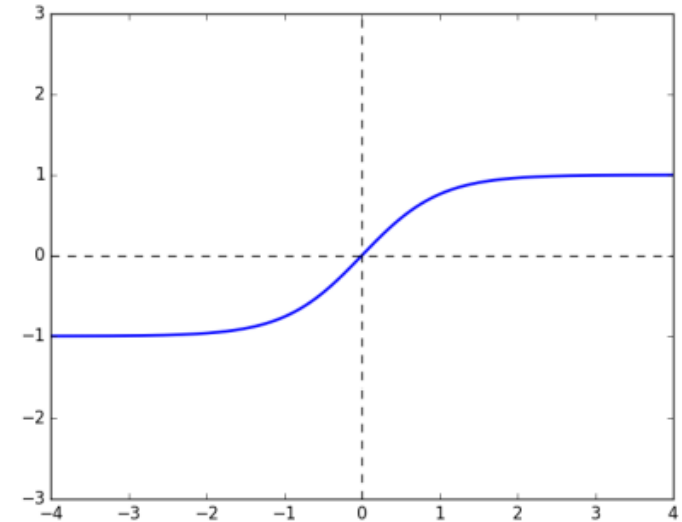
Hard Threshold

$$y = \begin{cases} 1 & \text{if } z > 0 \\ 0 & \text{if } z \leq 0 \end{cases}$$



Logistic

$$y = \frac{1}{1 + e^{-z}}$$



Hyperbolic Tangent (tanh)

$$y = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

Image from: [link](#)

Design choices in a multilayer network

- We have a supervised learning task (regression or classification), and we want to use a neural network. What are the design choices involved?

Architecture design:

1. **Input Layer:** These are the feature inputs. We can choose to add or remove the intercept/bias term.
2. **Hidden Layers:** The number of hidden layers to use. The number of hidden units in each hidden layer (these can be different per layer). The activation functions used (these can be different for each hidden unit). Whether or not to add an intercept/bias term for a hidden layer.
3. **Output Layer:** We can technically apply any activation for the output unit. Typically for regression a linear unit is used, and for classification a sigmoid unit is used.

Neural Networks are trainable graphs

- Neural networks can be thought of as directed acyclic graphs.
- The weights on the edges can be tuned (trained) to fit a dataset. This can be done using backpropagation technique (gradient descent).
- Why is it called deep learning? The name came from the success of neural networks with a lot of hidden layers in the 2000s.
- With this graph view, there is a lot of flexibility:
 - Fully-connected network (FCN): what we have seen so far.
 - Convolution neural networks (CNN): an architecture suitable for learning on image data (we will see later). Still relevant.
 - Recurrent neural networks (RNN): an architecture suitable for learning on temporal/ordered data like audio or text. Has somewhat lost popularity to transformers.
 - Transformers: an architecture for learning on temporal/ordered data. ChatGPT and language models use these. More efficient to train than RNNs.
 - Other graph architectures possible.

Natural Language Processing (NLP)

NLP Overview

- Natural Language Processing (NLP) is an AI field concerned with the interaction between computers and humans through natural language.
- The goal is to develop programs that can understand, interpret, and respond to human language in a way that is both meaningful and useful.
- Becoming integral part of everyday usage.
 - Applied in diverse fields like retail (e.g., customer service chatbots) and medicine (e.g., interpreting/summarizing electronic health records).
 - Used by conversational agents like Amazon's Alexa and Apple's Siri to understand and respond to user queries.
 - Develop advanced prose and power coherent chatbots like chatGPT.
 - Google uses NLP to enhance search engine results.
 - Social networks like Facebook use NLP to detect and filter hate speech.
- Nice textbook on NLP ([link](#)).

NLP Tasks

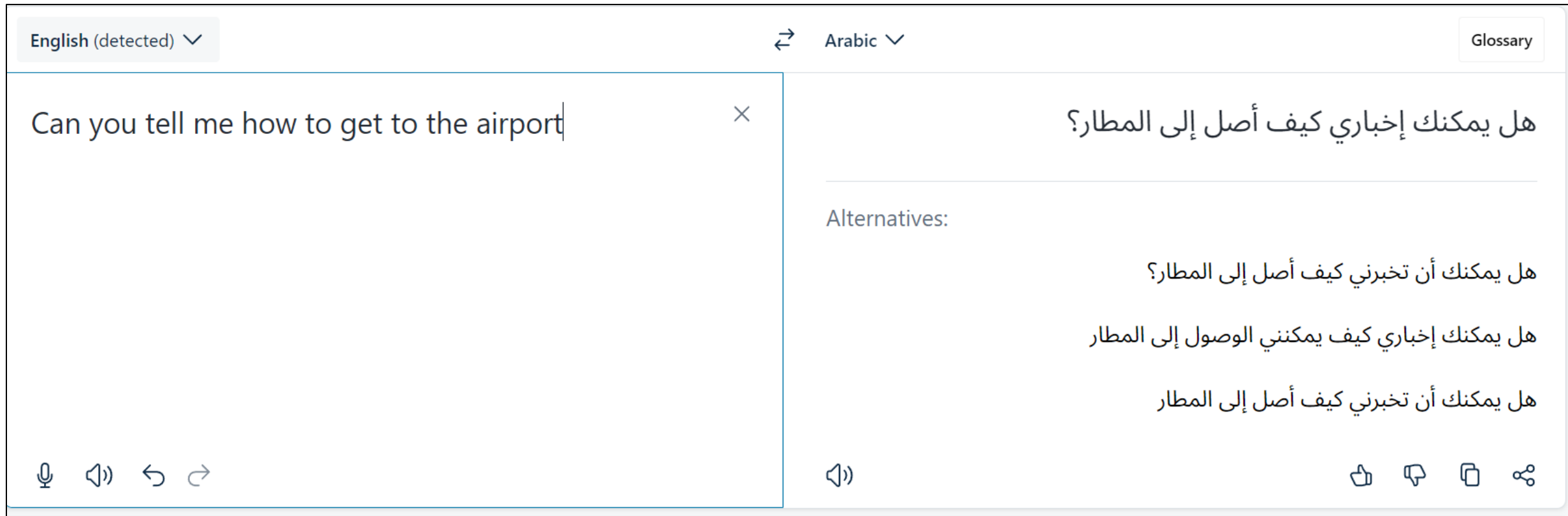
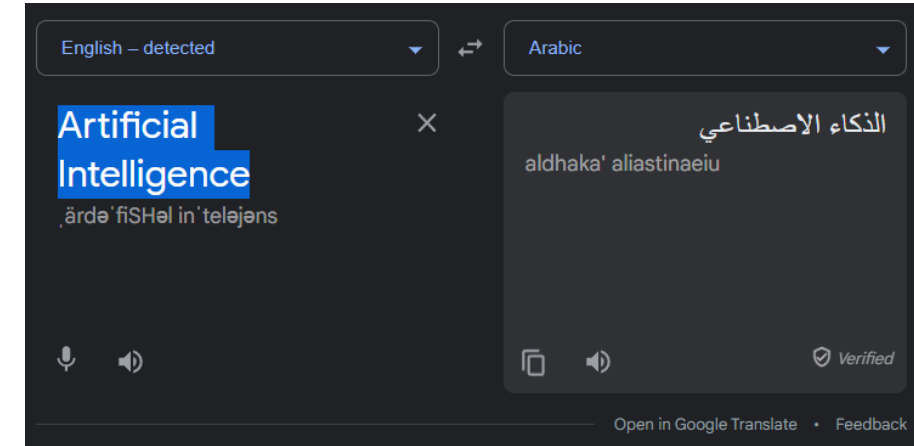
- Sentiment Analysis: given text, determine the emotional tone or sentiment expressed.



Figure from [link](#)

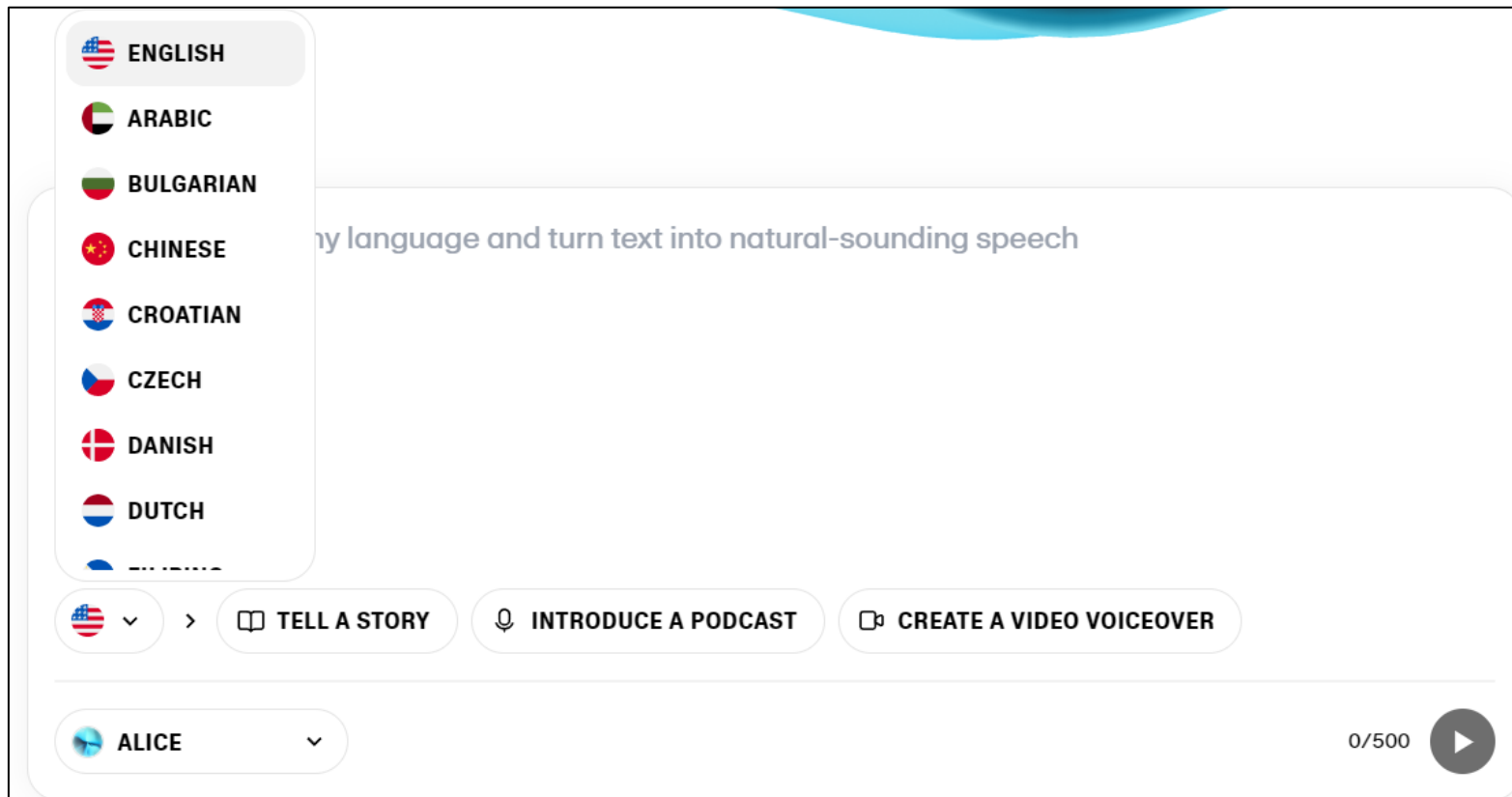
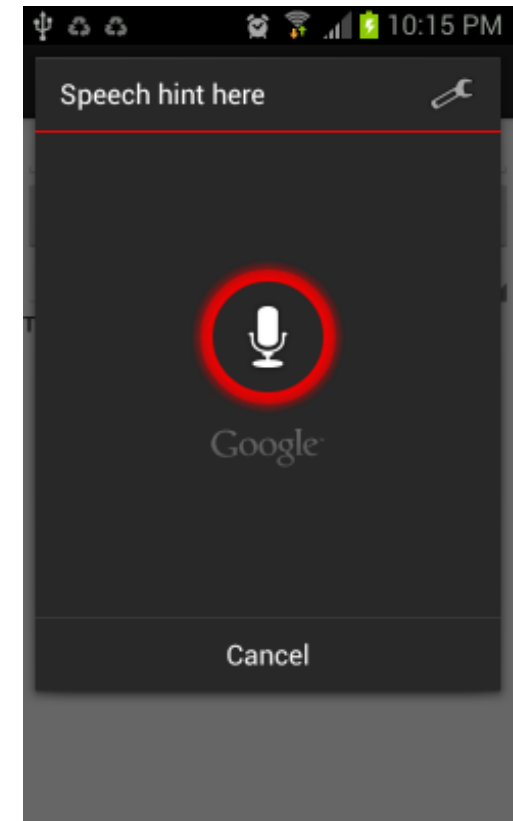
NLP Tasks

- Machine Translation: translate text from one language to another.



NLP Tasks

- Speech Recognition: convert spoken language into text.
- Text-to-Speech: convert text into spoken language.



<https://elevenlabs.io/text-to-speech>

NLP Tasks

- Question Answering: answer questions posed in natural language.
- Text Summarization: generate concise summaries of long documents.
- Language Generation: create human-like text based on input data.

You
Khalid is the son of Ahmed. Mohammed is the son of Ahmed. Waleed is the brother of Ahmed. What is the relationship between Mohammed and Waleed?

ChatGPT
Given the information provided:

- Khalid is the son of Ahmed.
- Mohammed is the son of Ahmed.
- Waleed is the brother of Ahmed.

Since both Mohammed and Khalid are sons of Ahmed, and Waleed is the brother of Ahmed, this implies that Mohammed and Waleed share the same father, making them brothers as well. Therefore, Mohammed and Waleed are brothers.



summarize this in bullet points:

The first documented computer architecture was in the correspondence between Charles Babbage and Ada Lovelace, describing the analytical engine. While building the computer Z1 in 1936, Konrad Zuse described in two patent applications for his future projects that machine instructions could be stored in the same storage used for data, i.e., the stored-program concept.[4][5] Two other early and important examples are:

John von Neumann's 1945 paper, First Draft of a Report on the EDVAC, which described an organization of logical elements;[6] and Alan Turing's more detailed Proposed Electronic Calculator for the Automatic Computing Engine, also 1945 and which cited John von Neumann's paper.[7]

The term "architecture" in computer literature can be traced to the work of Lyle R. Johnson and Frederick P. Brooks, Jr., members of the Machine Organization department in IBM's main research center in 1959. Johnson had the opportunity to write a proprietary research communication about the Stretch, an IBM-developed supercomputer for Los Alamos National Laboratory (at the time known as Los Alamos Scientific Laboratory) To describe the level of detail for discussing the luxuriously embellished computer, he noted that his description of

[ChatGPT](#)

NLP Current Modelling Trend

- The current successful trend in NLP is the use of deep learning ML models (supervised learning).
- However, ML models take in numbers as input.
- What is the overall architecture?
 - Input Tokenization: text is broken down into *tokens*.
 - Input Embedding: each token is converted to a numeric representation (many options here).
 - Deep learning model: typically uses transformer architecture.
 - Output: depends on task. If we want to predict text, then we can use output embeddings -> tokens -> text.

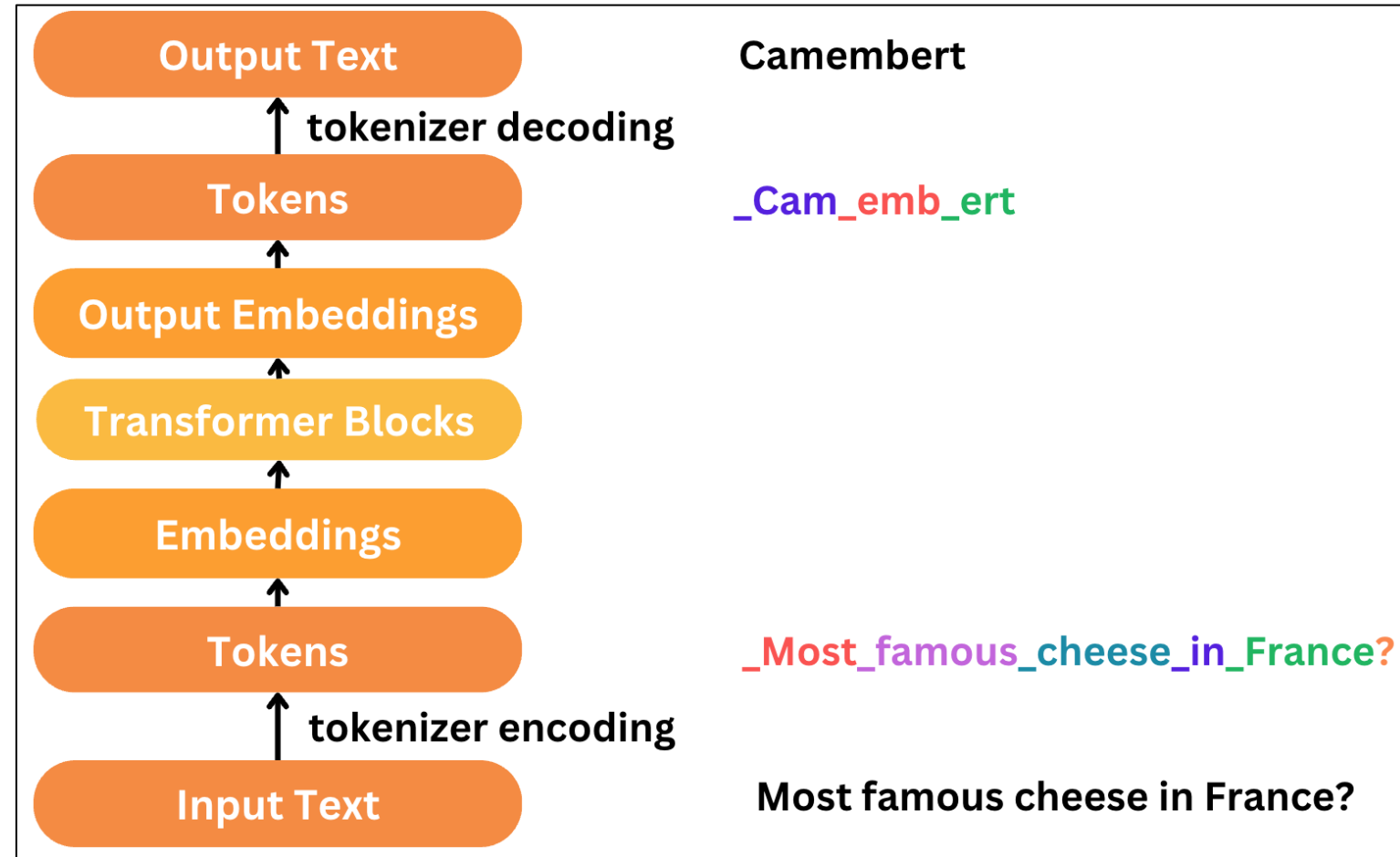
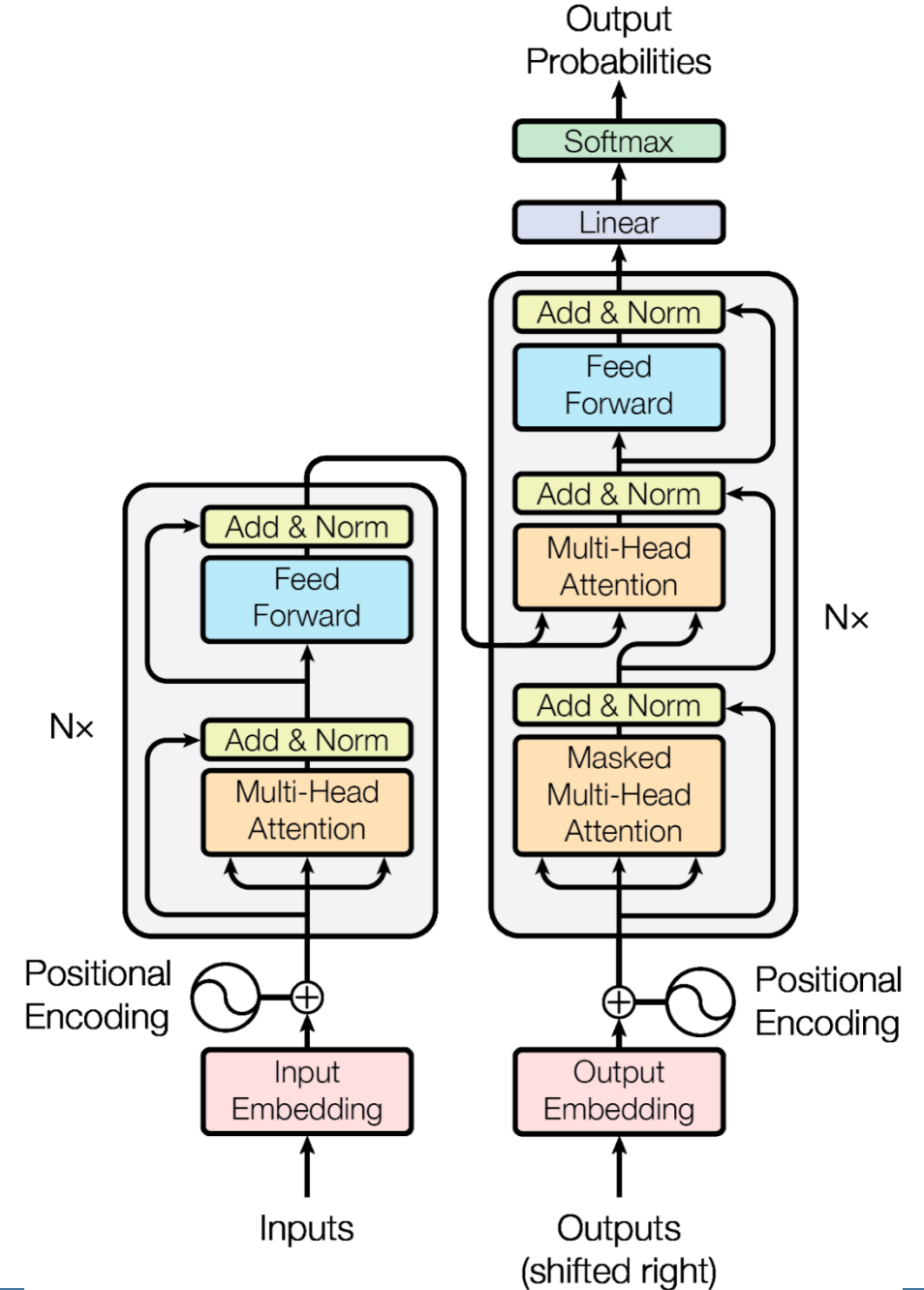


Figure from [link](#)

NLP Current LLM Trend

- For Large-language-models (LLMs) like ChatGPT, it uses transformer architecture.
- Has two parts:
 - Encoding layer: takes in the *context* (this is the text you give it).
 - Decoding layer: this is what does the text prediction. It does this in a sequential manner; predicts first word, then second, etc.
- [Video](#) giving a nice visualization of predicting text using this architecture.



NLP Current LLM Trend

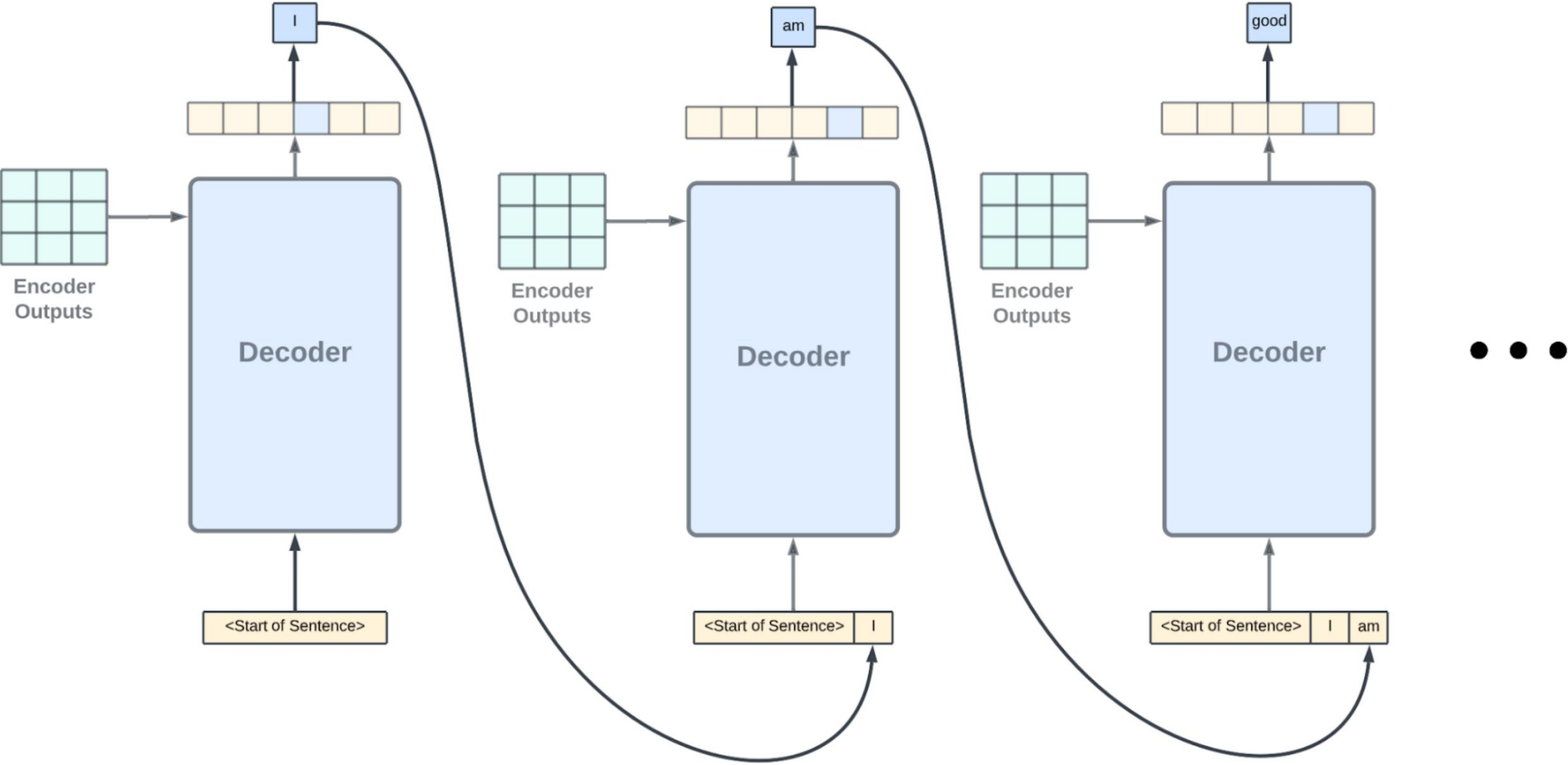


Figure from [link](#)

Robotics

Robotics Overview

- Multidisciplinary field integrating various branches of engineering and science (mechanical, electrical, control, computer science, AI) to design, build, and operate robots.
- Rough overview of areas:
 - Mechanical Design: focuses on the physical construction of robots (actuators, sensors, etc).
 - Control Systems: develop algorithms and systems to control the movement and behavior of robots.
 - Perception and Sensing: enabling robots to perceive their environment through sensors. Includes computer vision, LIDAR, RADAR, and touch sensors.
 - AI and ML: learn from experience, make decisions, and perform complex tasks autonomously.
 - Human-Robot Interaction (HRI): studies how humans and robots interact. Aims to make robots more user-friendly and intuitive. Involves user interface design, communication methods, and safety protocols.

Robotics Tasks

- Object Manipulation: includes pick and place, assembly, and sorting.



Robotics Tasks

- Navigation: includes path planning (like we've covered), localization (determine current position), mapping (explore environment).

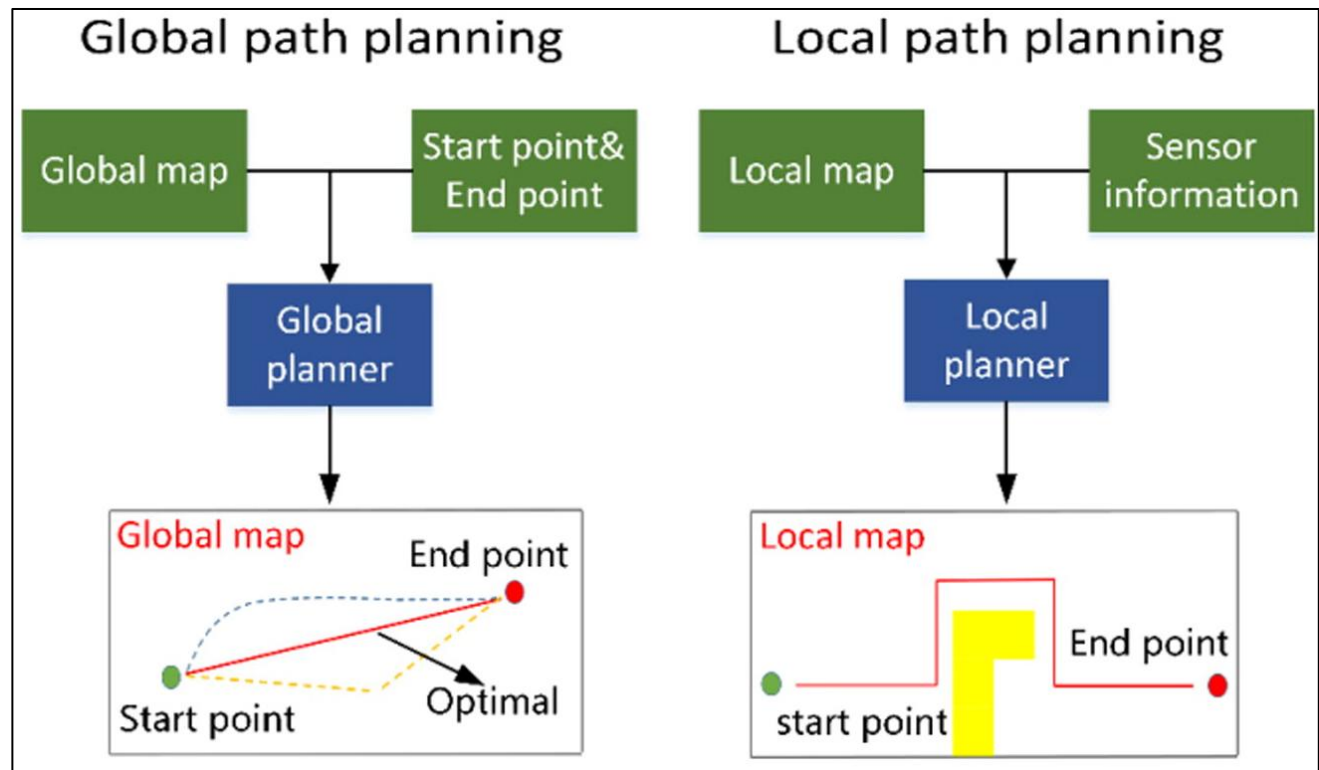


Figure from [link](#)

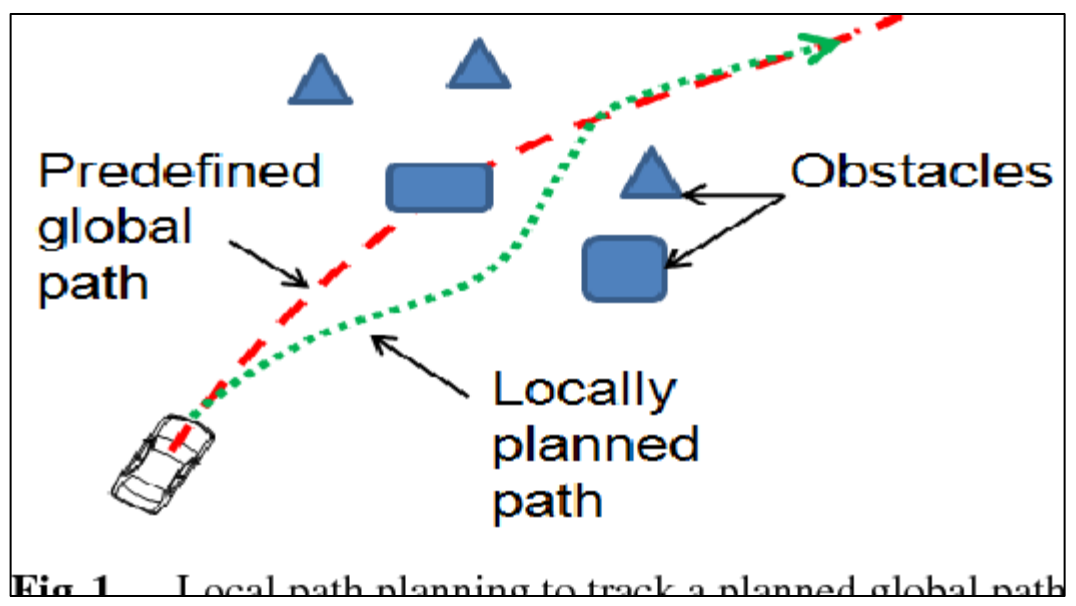
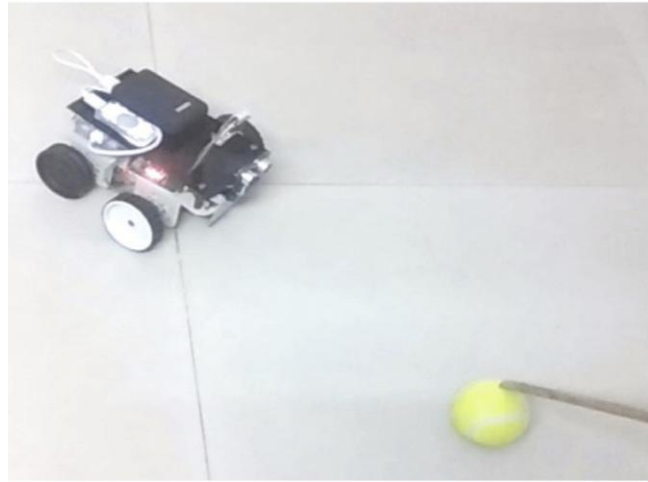


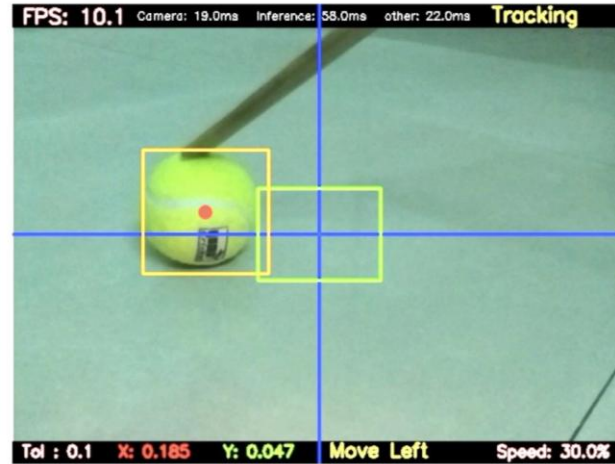
Figure from [link](#)

Robotics Tasks

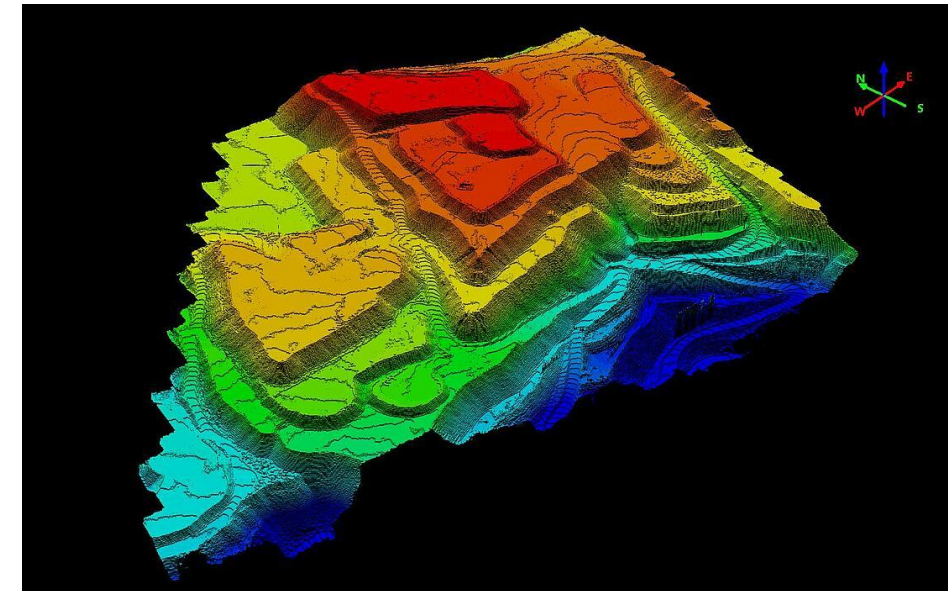
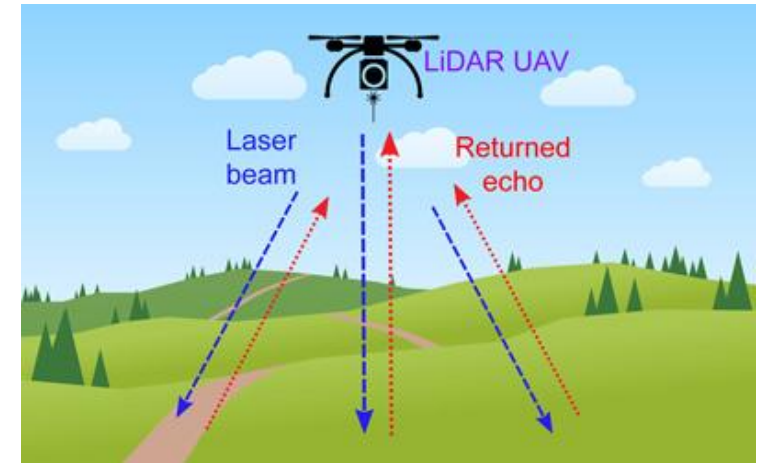
- Perception: includes object recognition, obstacle detection, and environment monitoring.



Object Tracking Robot



Robot's View



Figures from [link](#) and [link](#)

Figure from [link](#)

Robotics Tasks

- Human-Robot Interaction (HRI): speech recognition and generation, gesture recognition, intuitive displays and controls.

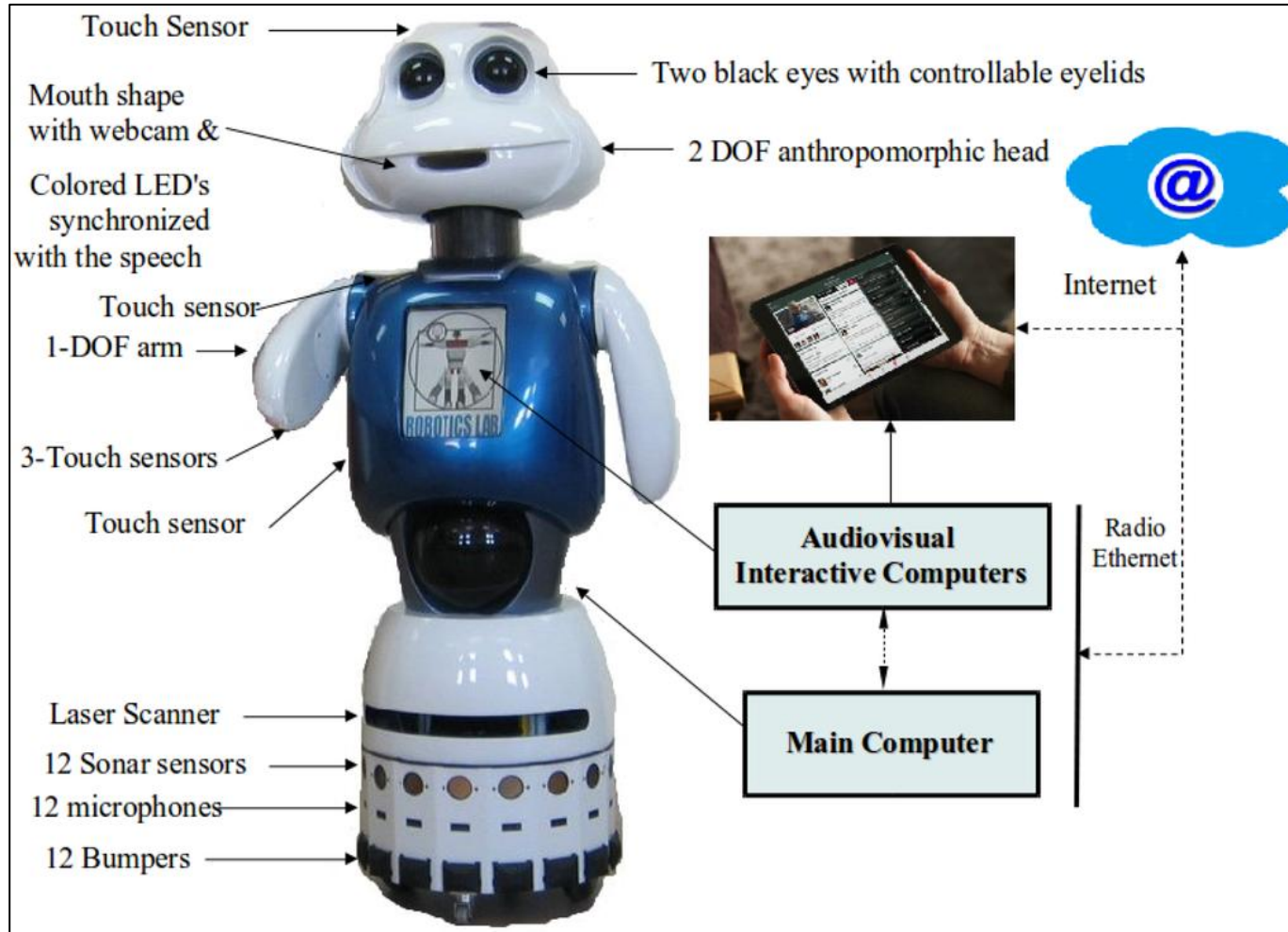


Figure from [link](#)

Robotics Tasks

- Autonomous Vehicles: cars, drones, planes, ships, space-ships. Delivery, exploration, monitoring, .

